

Fundamental Partial Differential Equations

Laplace and Poisson Equations

Abstract

This report introduces the Laplace and Poisson equations as the next mathematical step after two-dimensional convection, diffusion, and Burgers' equations. These elliptic equations are essential for understanding how pressure is computed in incompressible Navier--Stokes solvers. In particular, the pressure Poisson equation provides the mechanism by which the incompressibility constraint is enforced.

1. Motivation

In the previous stage, the focus was on extending one-dimensional model equations to two spatial dimensions. The main equations considered were two-dimensional convection, diffusion, and Burgers' equations.

These equations are time-dependent evolution equations. Given an initial condition, the solution is advanced forward in time.

For example, a general convection--diffusion equation may be written as

$$\frac{\partial u}{\partial t} + \text{convection} = \text{diffusion}.$$

However, the incompressible Navier--Stokes equations introduce an additional constraint:

$$\nabla \cdot \mathbf{u} = 0.$$

In two dimensions, this becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

This condition states that the velocity field must remain divergence-free. To enforce this constraint, pressure is obtained from an elliptic equation of the form

$$\nabla^2 p = b.$$

Therefore, before solving the full incompressible Navier--Stokes equations, it is necessary to understand how to solve Laplace and Poisson equations.

2. Evolution Equations versus Elliptic Equations

The equations studied previously, such as convection and diffusion, evolve in time. They require an initial condition and are advanced from one time level to the next.

For example, the diffusion equation is

$$\frac{\partial u}{\partial t} = \nu \nabla^2 u.$$

This equation describes how an initial field changes over time.

By contrast, elliptic equations such as the Laplace and Poisson equations do not describe time evolution directly. Instead, they describe a field determined by boundary conditions and, in the Poisson case, by internal source terms.

The Laplace equation is

$$\nabla^2 p = 0.$$

The Poisson equation is

$$\nabla^2 p = b.$$

These equations are global in character: the value of the solution at one point depends on the surrounding domain and the imposed boundary conditions.

3. The Laplace Equation

3.1 Governing Equation

The two-dimensional Laplace equation is

$$\nabla^2 p = 0,$$

or equivalently,

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = 0.$$

Here, $p = p(x, y)$ is the unknown scalar field.

3.2 Physical Interpretation

The Laplace equation appears in many steady-state problems, including:

- steady heat conduction,
- electrostatic potential,
- potential flow,
- steady pressure-like fields.

The key idea is that there are no internal sources or sinks. The solution is determined entirely by the boundary conditions.

3.3 Discretisation of the Laplace Equation

Let the computational domain be discretised using a uniform Cartesian grid:

$$x_i = i\Delta x, \quad y_j = j\Delta y.$$

The discrete approximation to $p(x_i, y_j)$ is denoted by

$$p_{i,j}.$$

Using central differences,

$$\frac{\partial^2 p}{\partial x^2} \approx \frac{p_{i+1,j} - 2p_{i,j} + p_{i-1,j}}{\Delta x^2},$$

and

$$\frac{\partial^2 p}{\partial y^2} \approx \frac{p_{i,j+1} - 2p_{i,j} + p_{i,j-1}}{\Delta y^2}.$$

Substituting into the Laplace equation gives

$$\frac{p_{i+1,j} - 2p_{i,j} + p_{i-1,j}}{\Delta x^2} + \frac{p_{i,j+1} - 2p_{i,j} + p_{i,j-1}}{\Delta y^2} = 0.$$

If the grid spacing is the same in both directions,

$$\Delta x = \Delta y,$$

then this simplifies to

$$p_{i,j} = \frac{1}{4} (p_{i+1,j} + p_{i-1,j} + p_{i,j+1} + p_{i,j-1}).$$

Thus, for a uniform grid, each interior point is updated as the average of its four nearest neighbours.

4. Iterative Solution of the Laplace Equation

The Laplace equation is usually solved iteratively. Starting from an initial guess, the interior values are repeatedly updated until the solution converges.

Let k denote the iteration index. A simple iterative update is

$$p_{i,j}^{k+1} = \frac{1}{4} (p_{i+1,j}^k + p_{i-1,j}^k + p_{i,j+1}^k + p_{i,j-1}^k).$$

The iteration continues until the change between successive approximations is sufficiently small:

$$\|p^{k+1} - p^k\| < \varepsilon,$$

where ε is a prescribed tolerance.

This convergence criterion ensures that the numerical solution has approached a steady state.

5. Boundary Conditions for the Laplace Equation

Because the Laplace equation is elliptic, boundary conditions play a central role.

A Dirichlet boundary condition prescribes the value of the solution:

$$p = g(x, y) \quad \text{on } \partial\Omega.$$

A Neumann boundary condition prescribes the normal derivative:

$$\frac{\partial p}{\partial n} = h(x, y) \quad \text{on } \partial\Omega.$$

Here, $\partial\Omega$ denotes the boundary of the domain.

The choice of boundary conditions determines the final solution. Unlike time-dependent equations, where the initial condition strongly influences the early evolution, elliptic equations are primarily boundary-driven.

6. The Poisson Equation

6.1 Governing Equation

The two-dimensional Poisson equation is

$$\nabla^2 p = b,$$

or

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = b(x, y).$$

The function $b(x, y)$ is called the source term.

6.2 Interpretation

The Poisson equation extends the Laplace equation by adding a source term. While the Laplace equation describes a field with no internal forcing, the Poisson equation describes a field influenced both by boundary conditions and by sources inside the domain.

If

$$b(x, y) = 0,$$

then the Poisson equation reduces to the Laplace equation.

7. Discretisation of the Poisson Equation

Using central differences, the Poisson equation becomes

$$\frac{p_{i+1,j} - 2p_{i,j} + p_{i-1,j}}{\Delta x^2} + \frac{p_{i,j+1} - 2p_{i,j} + p_{i,j-1}}{\Delta y^2} = b_{i,j}.$$

For equal grid spacing,

$$\Delta x = \Delta y,$$

the update equation is

$$p_{i,j} = \frac{1}{4} (p_{i+1,j} + p_{i-1,j} + p_{i,j+1} + p_{i,j-1} - b_{i,j} \Delta x^2).$$

This is similar to the Laplace update, except for the additional source contribution.

For unequal grid spacing, the update is

$$p_{i,j} = \frac{\Delta y^2 (p_{i+1,j} + p_{i-1,j}) + \Delta x^2 (p_{i,j+1} + p_{i,j-1}) - b_{i,j} \Delta x^2 \Delta y^2}{2 (\Delta x^2 + \Delta y^2)}.$$

8. Iterative Solution of the Poisson Equation

As with the Laplace equation, the Poisson equation is typically solved iteratively.

A generic iterative procedure is:

1. initialise the field p ,
2. impose boundary conditions,
3. update the interior points using the Poisson update formula,
4. reapply boundary conditions,
5. repeat until convergence.

The convergence condition may be written as

$$\|p^{k+1} - p^k\| < \varepsilon.$$

Alternatively, one may monitor the residual

$$r = \nabla^2 p - b,$$

and require

$$\|r\| < \varepsilon.$$

The residual measures how well the numerical approximation satisfies the Poisson equation.

9. Why the Poisson Equation Matters for Navier--Stokes

The incompressible Navier--Stokes equations are

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u},$$

with the incompressibility constraint

$$\nabla \cdot \mathbf{u} = 0.$$

In two dimensions, these become

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right), \end{aligned}$$

and

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

The pressure field is determined by enforcing the incompressibility condition. This leads to a pressure Poisson equation of the form

$$\nabla^2 p = b.$$

Thus, pressure is not simply another transported variable. It is the field that enforces the divergence-free condition on the velocity.

10. The Pressure Poisson Equation

In projection-type methods for incompressible flow, an intermediate velocity field is first computed without the pressure correction. This intermediate velocity is then corrected using pressure so that the final velocity field satisfies

$$\nabla \cdot \mathbf{u}^{n+1} = 0.$$

The pressure equation has the general form

$$\nabla^2 p = b,$$

where the right-hand side b depends on the velocity field.

In two dimensions, a commonly used form is

$$b = \rho \left[\frac{1}{\Delta t} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) - \left(\frac{\partial u}{\partial x} \right)^2 - 2 \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} - \left(\frac{\partial v}{\partial y} \right)^2 \right].$$

This term contains information about the local deformation and divergence of the velocity field.

The pressure Poisson equation is then solved at each time step before correcting the velocity field.

11. Top-Level Step-by-Step Guide for Week 4 Implementation

The implementation of the Laplace and Poisson solvers can be organised as follows.

Step 1: Define the Domain

Specify the rectangular domain

$$\Omega = [0, L_x] \times [0, L_y].$$

Choose the number of grid points

$$N_x, \quad N_y,$$

and compute

$$\Delta x = \frac{L_x}{N_x - 1}, \quad \Delta y = \frac{L_y}{N_y - 1}.$$

Step 2: Initialise the Field

For the Laplace or Poisson equation, initialise the scalar field

$$p_{i,j}^0.$$

This initial guess may be zero or some simple approximate field.

Step 3: Specify Boundary Conditions

Prescribe boundary conditions on

$$\partial\Omega.$$

These may be Dirichlet conditions,

$$p = g(x, y),$$

or Neumann conditions,

$$\frac{\partial p}{\partial n} = h(x, y).$$

Boundary conditions must be applied consistently throughout the iteration.

Step 4: Choose the Equation to Solve

For Laplace problems, solve

$$\nabla^2 p = 0.$$

For Poisson problems, solve

$$\nabla^2 p = b.$$

The Poisson equation requires specification of the source term

$$b(x, y).$$

Step 5: Iterate Over the Interior Points

Update each interior point using the appropriate finite-difference formula.

For the Laplace equation on a uniform grid:

$$p_{i,j}^{k+1} = \frac{1}{4} (p_{i+1,j}^k + p_{i-1,j}^k + p_{i,j+1}^k + p_{i,j-1}^k).$$

For the Poisson equation on a uniform grid:

$$p_{i,j}^{k+1} = \frac{1}{4} (p_{i+1,j}^k + p_{i-1,j}^k + p_{i,j+1}^k + p_{i,j-1}^k - b_{i,j} \Delta x^2).$$

Step 6: Reapply Boundary Conditions

After updating the interior values, reapply the boundary conditions.

This is essential because the boundary values determine the elliptic solution.

Step 7: Check Convergence

Measure the change between iterations:

$$\|p^{k+1} - p^k\|.$$

Stop the iteration when

$$\|p^{k+1} - p^k\| < \varepsilon.$$

Alternatively, monitor the residual

$$r = \nabla^2 p - b,$$

and stop when

$$\|r\| < \varepsilon.$$

Step 8: Interpret the Solution

For Laplace problems, interpret the solution as a steady field determined by the boundary conditions.

For Poisson problems, interpret the solution as a steady field determined by both boundary conditions and internal forcing.

In the Navier--Stokes context, interpret the solution as the pressure field required to enforce incompressibility.

12. Preparation for the Navier--Stokes Solver

The full incompressible Navier--Stokes solver will require the following sequence at each time step:

predict velocity $\rightarrow$ build pressure source term $\rightarrow$ solve pressure Poisson equation $\rightarrow$ correct velocity $\rightarrow$ apply boundary conditions

More explicitly:

1. compute the nonlinear convection terms,
2. compute the diffusion terms,
3. build the pressure source term b ,
4. solve

$$\nabla^2 p = b,$$

5. update the velocity using

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p,$$

6. enforce

$$\nabla \cdot \mathbf{u}^{n+1} \approx 0.$$

This is the conceptual bridge from two-dimensional Burgers' equations to incompressible Navier--Stokes.

13. Summary

This report introduced the mathematical and numerical role of the Laplace and Poisson equations in preparation for Navier--Stokes.

The key points are:

- the Laplace equation is

$$\nabla^2 p = 0,$$

- the Poisson equation is

$$\nabla^2 p = b,$$

- both are elliptic equations,
- their solutions are strongly determined by boundary conditions,
- they are usually solved iteratively,
- the pressure equation in incompressible flow is a Poisson equation,
- pressure enforces the divergence-free constraint,

$$\nabla \cdot \mathbf{u} = 0.$$

The next topic is the full incompressible Navier--Stokes solver, where convection, diffusion, pressure, and incompressibility are combined into a single computational algorithm.